

DECODING THE ROLES OF RHIZOBIA-INDUCED SOSEKI FAMILY MEMBERS IN CELL POLARITY PROCESSES DURING LEGUME ROOT SYMBIOSIS

Natasha HORTA ARAÚJO ^{1,*}, Anaïs DELERS ^{1,*}, Ambre GUILLORY ¹, Lisa FRANCES ¹,
Jean KELLER ², Joelle FOURNIER ¹, Fernanda DE CARVALHO-NIEBEL ¹

¹ Laboratory of Plant-Microbe-Environment Interactions (LIPME), Toulouse University, CNRS, INRAE, Castanet-Tolosan, France; ² Plant Sciences Research Laboratory (LRSV), Toulouse University, CNRS, Castanet-Tolosan, France; * These authors contributed equally to this work.

(corresponding authors: natasha.horta-araujo@inrae.fr; fernanda.de-carvalho-niebel@inrae)

Keywords: SOSEKI, Membrane, Polarity, Nitrogen-fixing legume Symbiosis

ABSTRACT

Legumes can establish beneficial symbiotic associations with rhizobial bacteria to acquire nitrogen, that is crucial for plant growth. This interaction leads to the formation of root nodules, specialized organs where bacteria fix atmospheric nitrogen for the host plant. Successful nodule development requires coordinated nodule organogenesis, driven by sequential reactivation of root cell divisions, and rhizobia entry towards the developing nodule, guided in most legumes by polarized infection threads¹. These processes rely on dynamic cellular polarization events, but the underlying molecular mechanisms are only beginning to emerge. Recent transcriptomic studies in the model legume *Medicago truncatula* identified two rhizobia-induced paralog genes encoding SOSEKI (SOK) plant cell polarity proteins, studied to date only in plant development². Here we show that these genes display tissue-specific expression patterns during rhizobial infection and nodule development. RNA interference and dominant-negative functional approaches in *M. truncatula* revealed that both SOKs are required for proper polarized infection and nodule development. Live-cell imaging studies in *Nicotiana benthamiana* and *M. truncatula* further uncovered dynamic, cell-specific SOK polarization patterns in rhizobia-infected and dividing root cells, with both intertwined and specialized subcellular localizations.

REFERENCES

- 1 F. de Carvalho-Niebel et al., *Curr Opin Plant Biol*, 81, 1-10 (2024), DOI: 10.1016/j.pbi.2024.102597
- 2 S. Yoshida, et al., *Nat Plants*, 5, 160-166, (2019), DOI: 10.1038/s41477-019-0363-6